

**Fireworks fly by**

Watch this amazing video of a drone capturing the Christmas fireworks
<http://dgit.in/DroneFireW>

**Jaasta keyboard**

A crowdfunded E-Ink keyboard that can morph its letters into shortcuts to match software you're using
<http://dgit.in/JaastaEInk>



Futurist Dr. Jordan Brandt on 4D Printing, the Indian Maker movement and more...

◀ Dr. Jordan Brandt,
 Manufacturing Technology
 Futurist at Autodesk

Siddharth Parwatay
siddharth.parwatay@digit.in

What does a "futurist" do exactly? Tell us about your Job at Autodesk

Jordan: I have an incredibly fun job, thinking about the future of designing and making things. At Autodesk historically we have been in the design and simulation space, and now we're quickly moving more into the manufacturing and construction side of things as well. I see all of these aspects converging, so my job is to look into this convergence, and that involves following a lot of trends. From robotics and automation to artificial intelligence, 3D printing, the Internet Of Things, all of these have influence on the way we design and make things.

Tell us more about distributed manufacturing systems, what exactly does that mean for regular folk?

Jordan: The idea is instead of having centralised large factories, we can actually distribute factories in a much smaller compact version and enable more local and regional production. An example of that would be Foxconn building their next largest factory in Indonesia. And it is not because the wages are cheaper or the regulatory environment is better, I mean those are factors, but the biggest reason is their next largest market is in Indonesia.

Also from the ground up you have things like the Maker Movement with increasing capacity for people to design and make their own things. You have 3D printing which gives you the technology to now design and make your own things. We see the convergence again from the large enterprise and from the individual maker communities coming together and all of this nets out at a more local design and production method.

Where do you see the Indian maker movement heading?

Jordan: Well, I think it's incredibly exciting now. This my first time in India, and just travelling around Mumbai for the last few days you see the resourcefulness of people. Whereas the wealth distribution seems to be skewed, the resourcefulness of the people is absolutely amazing. If you need something you can't afford to buy it, you figure out how to make it or how to modify something to enable you to solve whatever problem that you have at hand, you need to tap into that creative capacity, and a lot of that is access to tools. You don't basically need the formal education, that's what the internet's for right? You can learn the skill set you need. I would think that you guys are incredibly excited about the future of making in India, not to mention the fact that the needs of the people here are unique.

There's kind of been this truism since the end of the Second World War and Industrialization that industrialised economies produce a lot of the advanced products for the rest of the world. And the truth is that those industrialised economies probably don't know about the local needs in Mumbai, for example, so it makes sense for people to actually be empowered to make their own things.

We've been talking about 3D printing as the next big thing for years now, similar to the way people are talking about drones today but it hasn't become a household phenomenon yet. Will that ever happen?

Jordan: It will happen eventually, we have to give it time. I mean, you think of the typical hype cycle where we over-anticipate the growth and productisation of certain technologies, and about two years ago we hit the peak of the hype cycle around consumer 3D printing – that is going to be in every home and also these grand expectations. There's a lot of problems preventing that from happening today, one of them being the reliability of the hardware. Anyone who has a 3D printer knows that their reliability isn't there, the repeatability isn't there. Personally I have three 3D Printers, and two of them are used to make spare parts for the third one. Using them requires an Engineering or kind of troubleshooting mentality, which